



Department Of Information Technology
Android Development Framework | BETCE16330
Important Question Bank

Sr.no	Question
Unit 1 :- Introduction to Flutter & Dart Basics	
1	Explain Flutter framework and its advantages for cross-platform mobile application development.
2	Describe Flutter architecture with proper explanation of its main components.
3	Explain the concept of Widgets in Flutter and classify them.
4	Explain the steps to set up Flutter SDK and configure IDE for Flutter development.
5	Explain the role of Dart programming language in Flutter development.
6	Discuss variables and data types in Dart with suitable examples.
7	Explain functions in Dart and their importance in Flutter applications.
8	Explain object-oriented concepts supported by Dart language.
9	Describe Dart collections (List, Set, Map) with examples.
10	Explain the structure of a basic Flutter application.
Unit 2 :- Flutter UI Design with Widgets	
11	Explain the concept of Widget Tree in Flutter with a suitable example.
12	Differentiate between Stateless and Stateful widgets with example.
13	Explain Row and Column widgets and their properties.
14	Explain Container widget and its role in UI design.
15	Describe Stack widget and its use cases in Flutter UI.
16	Explain commonly used input widgets in Flutter.
17	Explain TextField widget and its properties.
18	Explain Button widgets available in Flutter.
19	Explain theming and styling in Flutter applications.
20	Explain responsive UI design in Flutter for multiple screen sizes.
Unit 3 :- State Management and Navigation	
21	Explain the concept of state in Flutter applications.
22	Explain setState() method and its role in state management.
23	Differentiate between Stateless widget state and Stateful widget state.
24	Explain the lifecycle of a Stateful widget.

25	What is InheritedWidget? Explain its purpose in Flutter.
26	Explain Provider package and its importance in state management.
27	Explain navigation in Flutter using routes.
28	Explain named routes and their advantages.
29	Explain passing data between screens in Flutter.
30	Explain form handling and validation in Flutter.
Unit 4 :- Data Handling and Integration	
31	Explain JSON data format and its role in Flutter applications.
32	Explain RESTful APIs and their use in mobile applications.
33	Explain the steps to fetch data from an API in Flutter.
34	Explain JSON parsing in Flutter with example.
35	Explain ListView widget and its importance in displaying dynamic data.
36	Explain GridView widget and its use cases.
37	Explain local data storage using SharedPreferences.
38	Explain SQLite database and CRUD operations in Flutter.
39	Explain Hive database and its advantages over SQLite.
40	Give an overview of Firebase integration in Flutter applications.
Unit 5 :- Advanced Features & Deployment	
41	Explain animations in Flutter and their importance in user experience.
42	Explain custom widgets and their advantages in Flutter applications.
43	Explain gesture detection in Flutter.
44	Explain how Flutter accesses device features such as camera and location.
45	Explain the use of third-party packages in Flutter.
46	Explain the role of pub.dev in Flutter development.
47	Explain debugging techniques in Flutter applications.
48	Explain testing in Flutter and its types.
49	Explain the process of deploying a Flutter app to Play Store.
50	Explain version control using Git and GitHub for Flutter projects.





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General Question Bank

Sr.no	Question
Unit 1 :- Introduction to Flutter & Dart Basics	
1	What is Flutter and why is it preferred over traditional native development?
2	Explain cross-platform application development with reference to Flutter.
3	Describe the layered architecture of Flutter framework.
4	What are widgets in Flutter? Explain their role in UI design.
5	Differentiate between Flutter and native Android development.
6	Explain the process of installing Flutter SDK on a development system.
7	Discuss the importance of IDEs like Android Studio or VS Code in Flutter development.
8	Explain Dart language features that make it suitable for Flutter.
9	Describe different types of variables supported by Dart.
10	Explain control flow statements used in Dart programming.
11	What are functions in Dart? Explain with syntax and example.
12	Explain classes and objects in Dart with suitable example.
13	Describe constructors in Dart programming.
14	Explain inheritance and polymorphism concepts in Dart.
15	What are Dart collections? Explain List collection with example.
16	Explain Map collection in Dart with real-world use case.
17	Explain Set collection and its advantages.
18	Describe the basic structure of main() function in Flutter.
19	Explain how Flutter applications are executed on emulator or real device.
20	Write a short note on Hot Reload and its advantages in Flutter development.
Unit 2 :- Flutter UI Design with Widgets	
21	What is a widget tree? Explain its importance in Flutter UI rendering.
22	Explain Stateless widget with real-time example.
23	Explain Stateful widget and justify its need in Flutter applications.
24	Compare Stateless and Stateful widgets based on lifecycle and usage.
25	Explain Row widget with alignment and spacing properties.
26	Explain Column widget and its layout behavior.
27	Explain Container widget properties such as padding, margin, and decoration.

28	What is Stack widget? Explain how it differs from Row and Column.
29	Explain positioning of widgets using Stack and Positioned widget.
30	Explain Text widget and its styling properties.
31	Explain TextField widget and user input handling.
32	Explain different types of button widgets used in Flutter.
33	Explain Checkbox and Switch widgets with example.
34	Explain Radio button widget and its use case.
35	Explain theme data and global styling in Flutter.
36	What is responsive design? Explain its importance in Flutter apps.
37	Explain MediaQuery and its role in responsive UI design.
38	Explain layout widgets used for building complex UI structures.
39	Explain how Flutter adapts UI for different device orientations.
40	Write a short note on best practices for Flutter UI design.
Unit 3 :- State Management and Navigation	
41	What is state in Flutter? Why is state management required?
42	Explain the difference between local state and global state.
43	Explain how setState() updates the UI in Flutter.
44	Describe the lifecycle methods of a StatefulWidget.
45	How do you manage different screen sizes and densities in your Android app?
46	Explain initState() and dispose() with use cases.
47	Why is InheritedWidget considered inefficient for large applications?
48	Explain Provider architecture and its working principle.
49	Compare setState() and Provider for state management.
50	Explain navigation stack in Flutter.
51	Explain Navigator.push() and Navigator.pop() methods.
52	Explain named routes with syntax and example.
53	Explain passing arguments between routes.
54	Explain Bottom Navigation in Flutter applications.
55	Explain Drawer widget and its role in app navigation.
56	Explain form widget and its importance in Flutter.
57	Explain form validation techniques in Flutter.
58	Explain TextFormField and validator property.
59	Explain global keys and their use in form validation.
60	Write a short note on best practices for state management in Flutter.
Unit 4 :- Data Handling and Integration	
61	What is JSON? Explain its structure with example.
62	Explain REST architecture and HTTP methods used in APIs.
63	Explain how Flutter communicates with RESTful web services.
64	Describe asynchronous programming used during API calls in Flutter.
65	Explain the role of http package in Flutter.
66	Explain JSON decoding and encoding in Dart.
67	Explain ListView.builder and its advantages.
68	Compare ListView and GridView widgets.
69	Explain pagination while loading large datasets in Flutter.
70	Explain SharedPreferences for key-value data storage.
71	Explain use cases of SharedPreferences in mobile apps.
72	Explain SQLite database architecture in Flutter.

73	Explain CRUD operations using SQLite.
74	Compare SQLite and Hive databases.
75	Explain advantages of Hive for Flutter applications.
76	Explain data persistence and its importance in mobile apps.
77	Explain basic Firebase services used with Flutter.
78	Explain Firebase Authentication in brief.
79	Explain Firebase Cloud Firestore overview.
80	Write a short note on best practices for handling API data in Flutter.
Unit 5 :- Advanced Features & Deployment	
81	What are animations in Flutter? Explain different animation types.
82	Explain implicit and explicit animations in Flutter.
83	Explain custom widget creation with example.
84	Why are custom widgets important for large Flutter applications?
85	Explain gesture detection using GestureDetector widget.
86	Explain handling user interactions in Flutter applications.
87	Explain accessing camera in Flutter applications.
88	Explain accessing location and sensor data in Flutter.
89	Explain the need for third-party packages in Flutter.
90	Explain package management in Flutter using pubspec.yaml.
91	Explain debugging tools available in Flutter.
92	Explain Flutter DevTools and its features.
93	Explain different testing approaches in Flutter.
94	Explain unit testing in Flutter.
95	Explain widget testing in Flutter.
96	Explain integration testing in Flutter.
97	Explain the steps involved in publishing an app to Play Store.
98	Explain the process of generating signed APK/IPA files.
99	Explain Git repository creation and code management workflow.
100	Write a short note on collaborative development using GitHub.



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Twisted Question Bank

Sr.no	Question
Unit 1 :- Introduction to Flutter & Dart Basics	
1	Why is Flutter considered both a UI framework and a complete SDK? Explain with architectural justification.
2	How do Dart's object-oriented features contribute to Flutter's widget-based architecture?
3	Compare Dart collections and justify which collection is best suited for managing dynamic UI data in Flutter apps.
4	Analyze how Flutter architecture improves performance compared to other cross-platform frameworks.
5	Suppose you are developing a large Flutter application—how would Dart language features help in maintaining scalability and code reusability?
6	Flutter claims “everything is a widget.” Analyze this statement by linking Flutter architecture, Dart classes, and UI rendering mechanism.
7	If Dart did not support object-oriented programming, how would it impact Flutter's widget tree and application structure? Explain logically.
8	You are asked to build a scalable Flutter application. Analyze how Dart collections and functions together help in managing UI components efficiently.
9	Evaluate the role of Hot Reload in Flutter development. How does it internally interact with Dart code execution and widget rebuilding?
10	Analyze the relationship between Flutter SDK, Dart language, and IDE tools in the complete app development lifecycle.
Unit 2 :- Flutter UI Design with Widgets	
11	Analyze how the widget tree structure affects UI rendering performance in Flutter.
12	Why are Stateful widgets not always preferred? Analyze with real-world UI scenarios.
13	Compare Row, Column, and Stack widgets and justify where each should be used in complex UI design.
14	How does Flutter manage layout constraints internally while rendering widgets?
15	Analyze how improper widget nesting can impact UI performance and readability.
16	Design a responsive Flutter UI layout for multiple screen sizes and justify your widget choices.

17	Evaluate the role of themes in maintaining UI consistency across large Flutter applications.
18	How do input widgets interact with state management in real-time user interfaces?
19	Analyze how Flutter handles UI updates when widget properties change dynamically.
20	A Flutter UI works well on mobile but fails on tablet devices. Analyze the possible design mistakes and suggest solutions.
Unit 3 :- State Management and Navigation	
21	Analyze why improper state management leads to performance issues in Flutter applications.
22	Compare setState(), InheritedWidget, and Provider in terms of scalability and performance.
23	A Flutter app works correctly initially but slows down as screens increase. Analyze possible state management mistakes.
24	How does Flutter's navigation stack interact with widget lifecycle methods? Explain logically.
25	Analyze how passing data between screens can affect app architecture in large Flutter projects.
26	Design a navigation flow using Bottom Navigation and Drawer together. Justify your approach.
27	Evaluate the role of Provider in separating UI and business logic.
28	How does form validation interact with state management internally in Flutter?
29	Analyze a scenario where improper use of setState() causes unnecessary widget rebuilds.
30	You are developing a multi-screen Flutter app with shared data. Analyze and justify the best state management approach.
Unit 4 :- Data Handling and Integration	
31	Analyze how improper API handling can affect Flutter app performance and user experience.
32	Compare SharedPreferences, SQLite, and Hive and justify which should be used in different real-world scenarios.
33	A Flutter app crashes while parsing JSON data from an API. Analyze possible causes and solutions.
34	How does asynchronous API calling interact with widget rebuilding in Flutter? Explain logically.
35	Analyze the challenges of displaying large datasets using ListView and GridView.
36	Design a Flutter application architecture that integrates API data with local storage.
37	Evaluate the role of Firebase in building scalable Flutter applications.
38	Analyze security concerns while consuming REST APIs in Flutter apps.
39	A Flutter app must work both online and offline. Analyze how data handling should be designed.
40	How does data persistence improve app reliability and performance? Analyze with justification.
Unit 5 :- Data Handling and Integration	
41	Analyze how animations and gestures together enhance user experience in Flutter applications.

42	Evaluate the impact of poor testing practices on Flutter app reliability.
43	A Flutter app performs well locally but fails after deployment. Analyze possible causes.
44	How does using third-party packages affect app performance and security? Analyze critically.
45	Design a Flutter app architecture that includes custom widgets, animations, and device features.
46	Analyze how Flutter DevTools assists in debugging complex UI and performance issues.
47	Compare unit testing, widget testing, and integration testing in real-world Flutter projects.
48	Analyze the challenges involved in publishing Flutter apps on Play Store and App Store.
49	Evaluate the role of GitHub in collaborative Flutter development and version control.
50	A Flutter project involves multiple developers and frequent updates. Analyze best practices to avoid deployment failures.

